Lab HA/BCP — Session B: back up, restore, write the recovery plan

GRETA de l'Essonne — AIS professional title — Block 2: Design and evolution

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Charles Maduba — Group 3 (work carried out in a team of 4 trainees)

Session B: Monday 12 October 2026. Follows session A (cluster, NFS, HA, RTO = 2 min 17 s).

English version of the lab report, for personal reference. The original report is written in French.

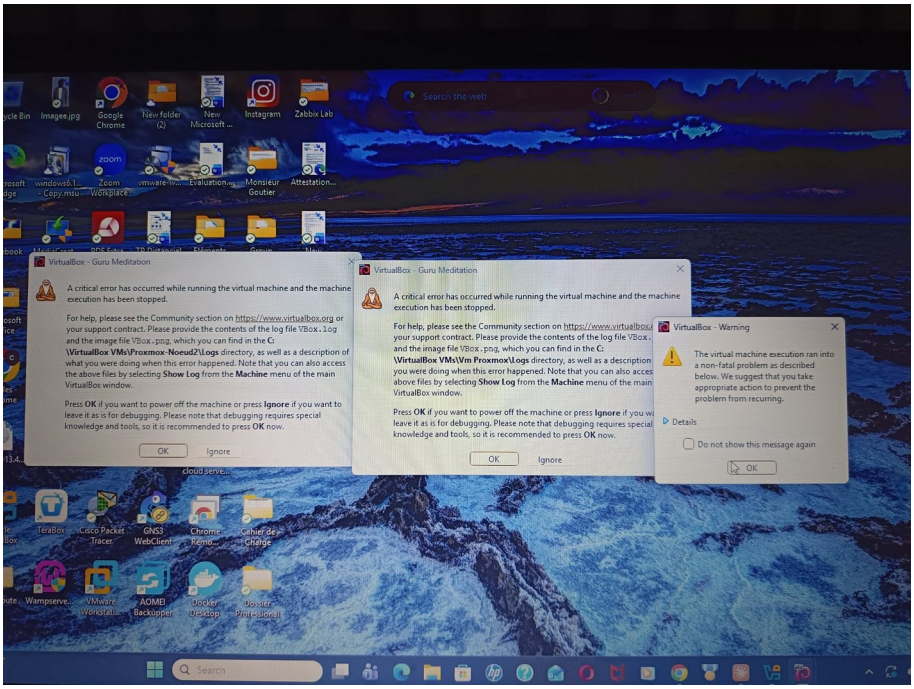
Summary

Part	Content	Result
0	Catch-up: cluster, HA, shared disk	Checked
1	Backup, restore, automatic backup	Backup 10 s, restore 3.6 s, daily job at 21:00 (keep 7)
2	RPO	Estimated per scenario (VM without an OS)
3	Disaster recovery plan (DRP)	Written (6 sections)
4	Cross-group presentation	Group 3 values ready

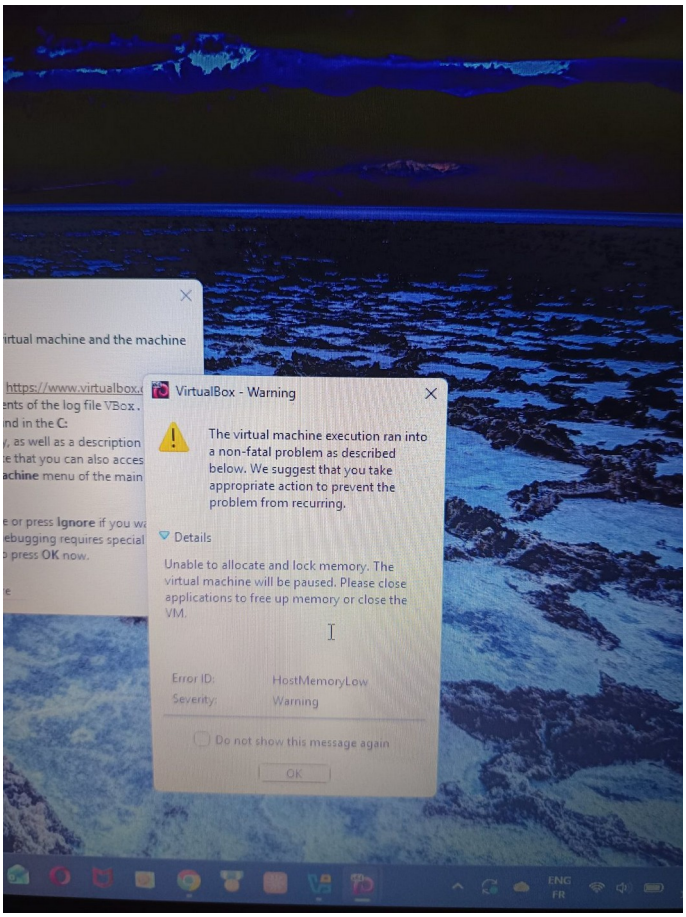
All operations were done on the command line (vzdump, qmrestore, pvesh): with the 3 nodes running, the host PC does not have enough memory to open the web interface. These commands are the documented equivalents of the web interface buttons (the lab guide lists them).

Start-up incident: not enough memory on the host PC

On the first start of the 3 nodes, VirtualBox stopped pve and pve2 ("Guru Meditation") with the error HostMemoryLow: "Unable to allocate and lock memory". The 3 nodes need 6 GB of RAM on an 11.8 GB PC where Chrome, the Claude app and WhatsApp were also running. After closing these applications, the 3 nodes started normally.



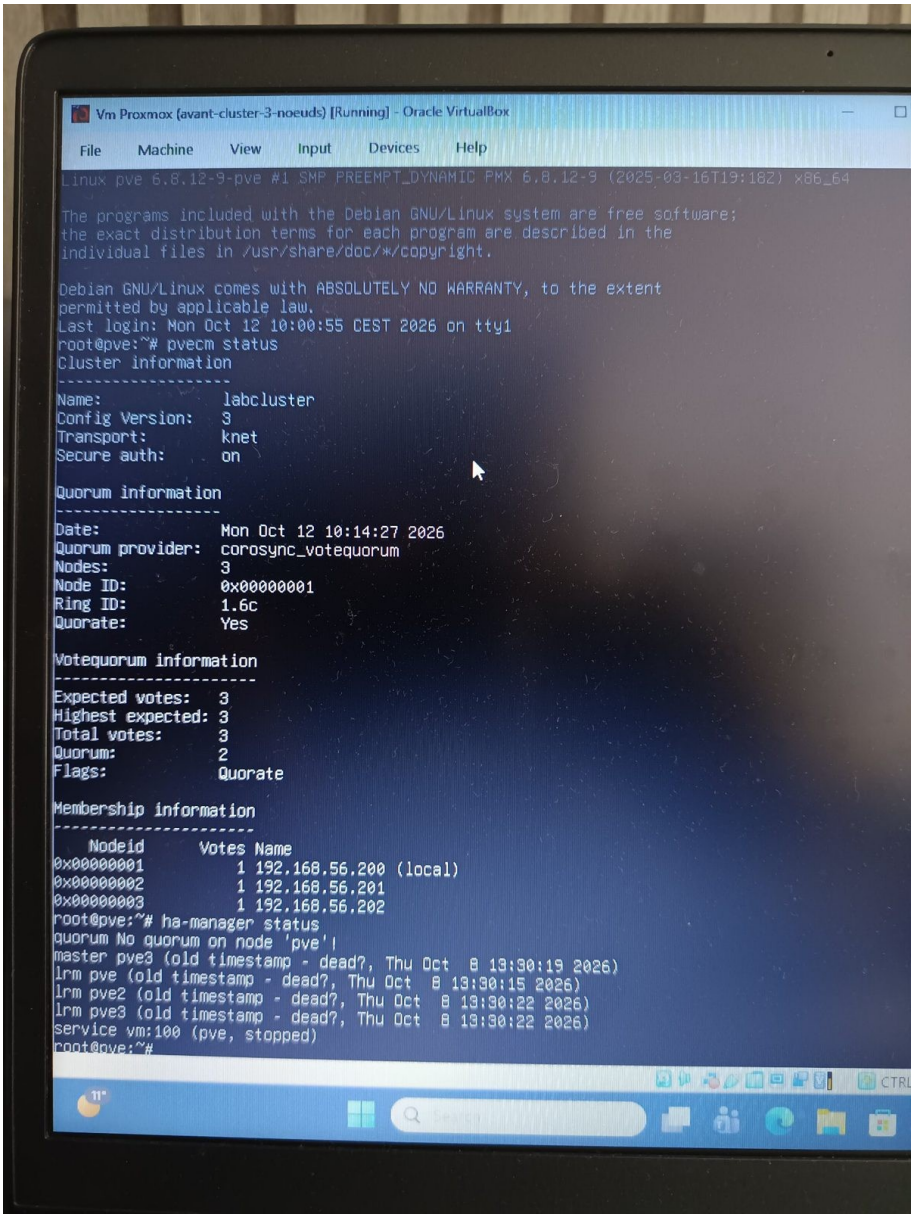
pve and pve2 stopped by VirtualBox (Guru Meditation)



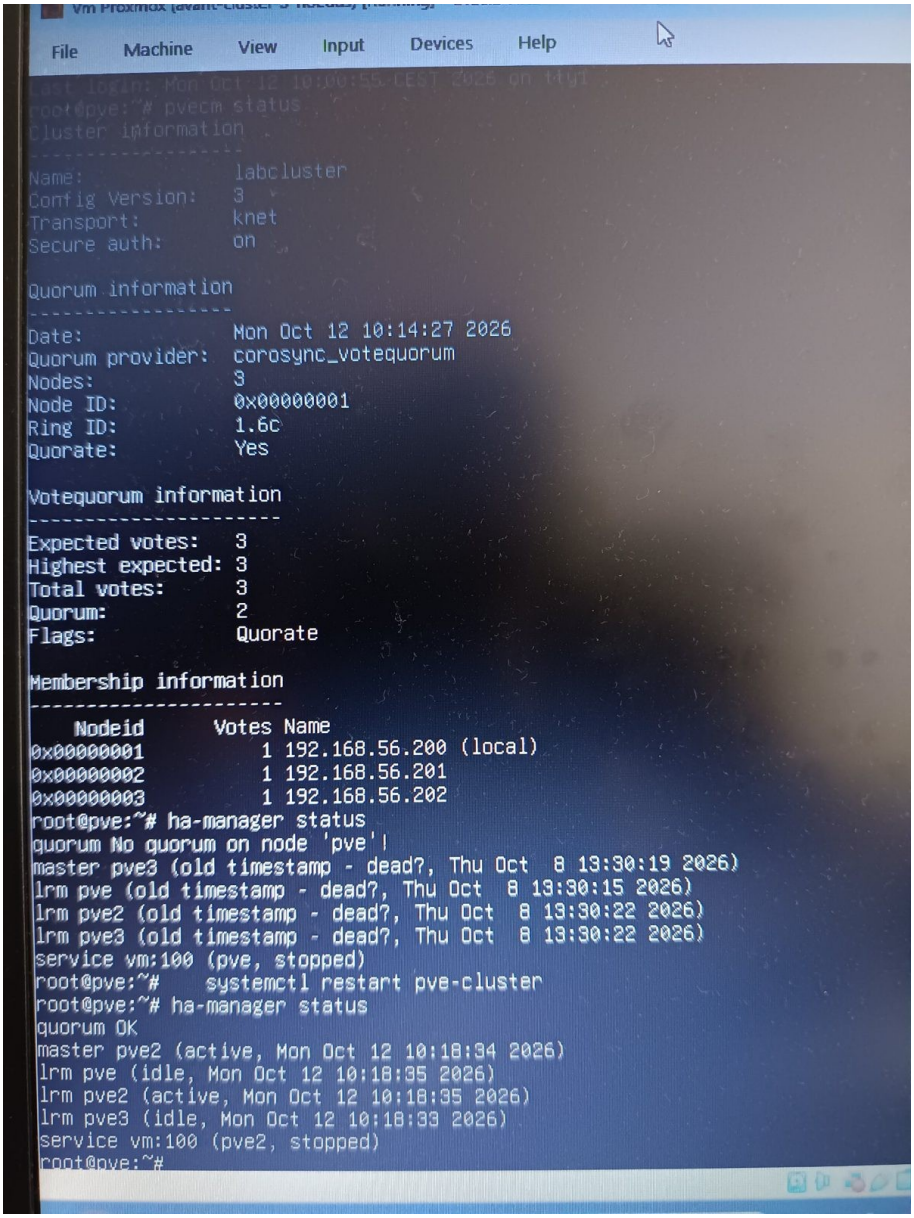
Error detail: HostMemoryLow

PART 0 — Checking session A

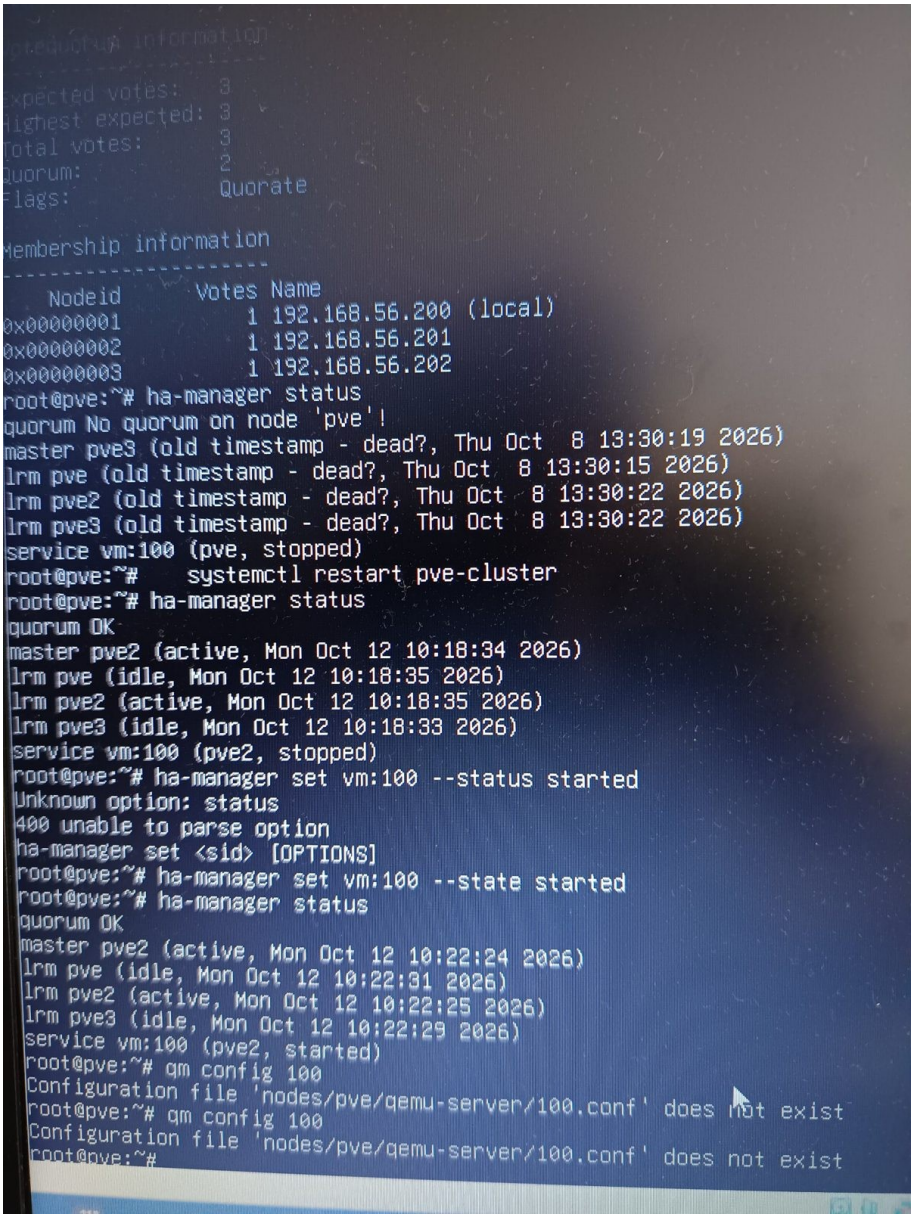
pvecm status showed 3 votes and Quorate: Yes, but ha-manager status showed "No quorum on node 'pve'" with timestamps from 8 October: the cluster configuration service (pmxcfs) had stayed stuck at start-up. Fixed with systemctl restart pve-cluster (same problem as in session A). HA then took over again and placed VM-Test on pve2.



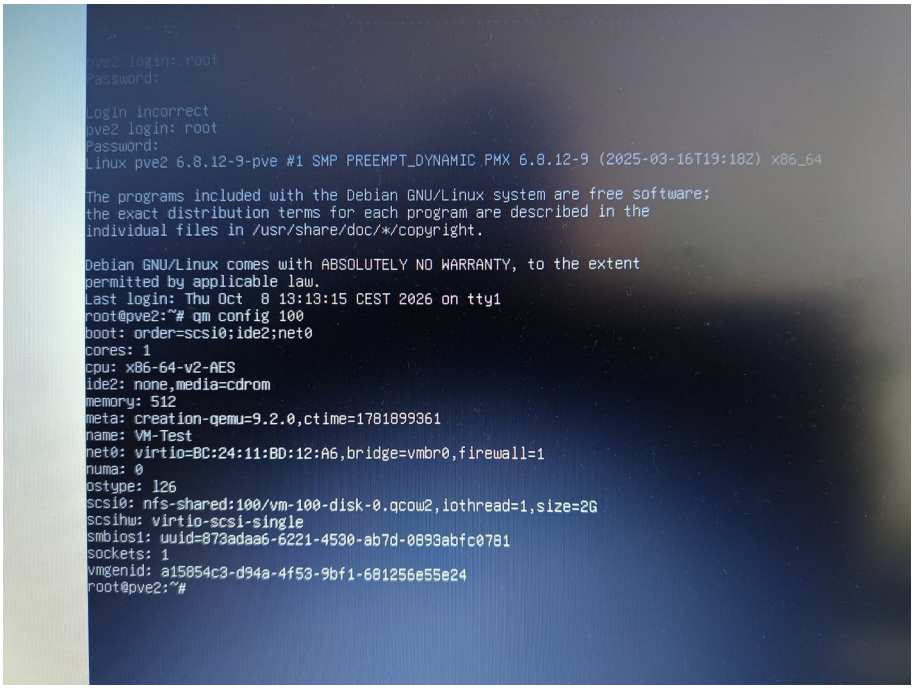
pvecm status: 3 votes, Quorate: Yes; ha-manager: timestamps from 8 October



After systemctl restart pve-cluster: quorum OK, VM 100 on pve2



ha-manager set vm:100 --state started → service vm:100 (pve2, started)

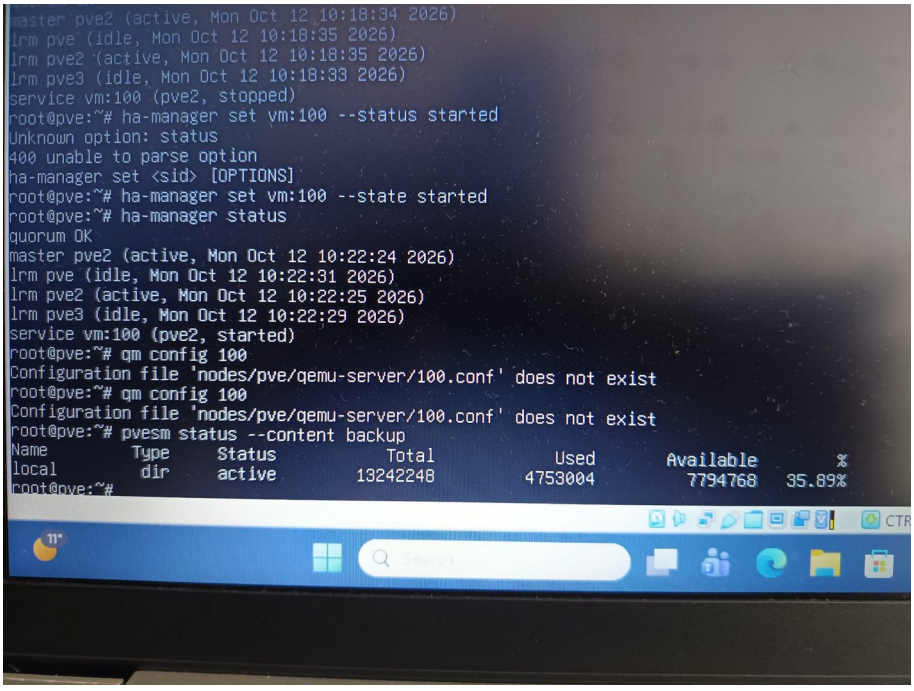


qm config 100 on pve2: scsi0 disk on nfs-shared, 2 GB

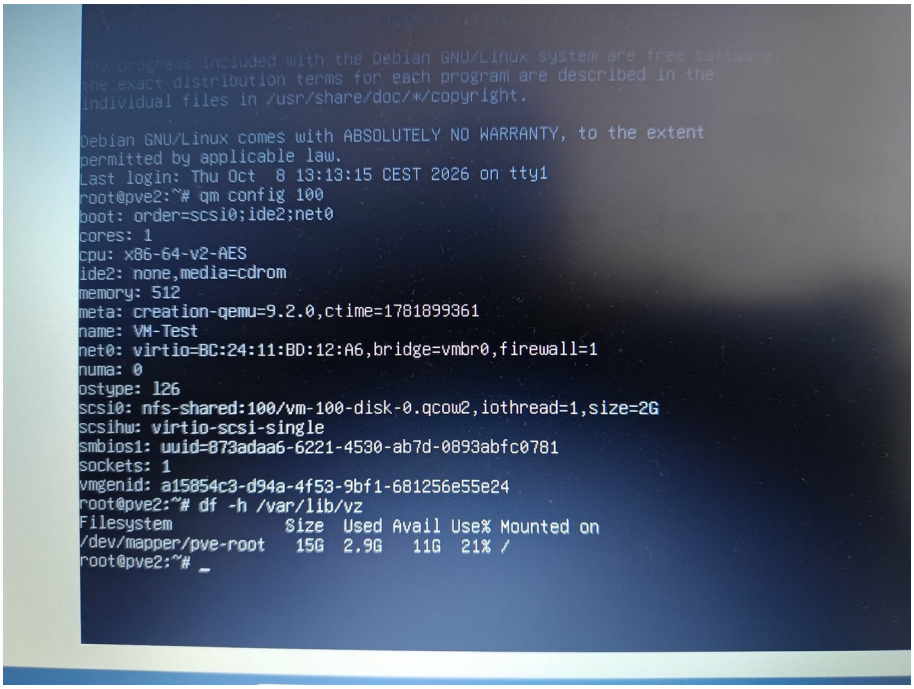
PART 1 — Back up and restore

1.1 — Where to store the backups

The local storage (/var/lib/vz) accepts backups. On pve2 (the node running the VM), 11 GB are free, far more than VM-Test's 2 GB disk.



pvesm status --content backup: local, dir, active

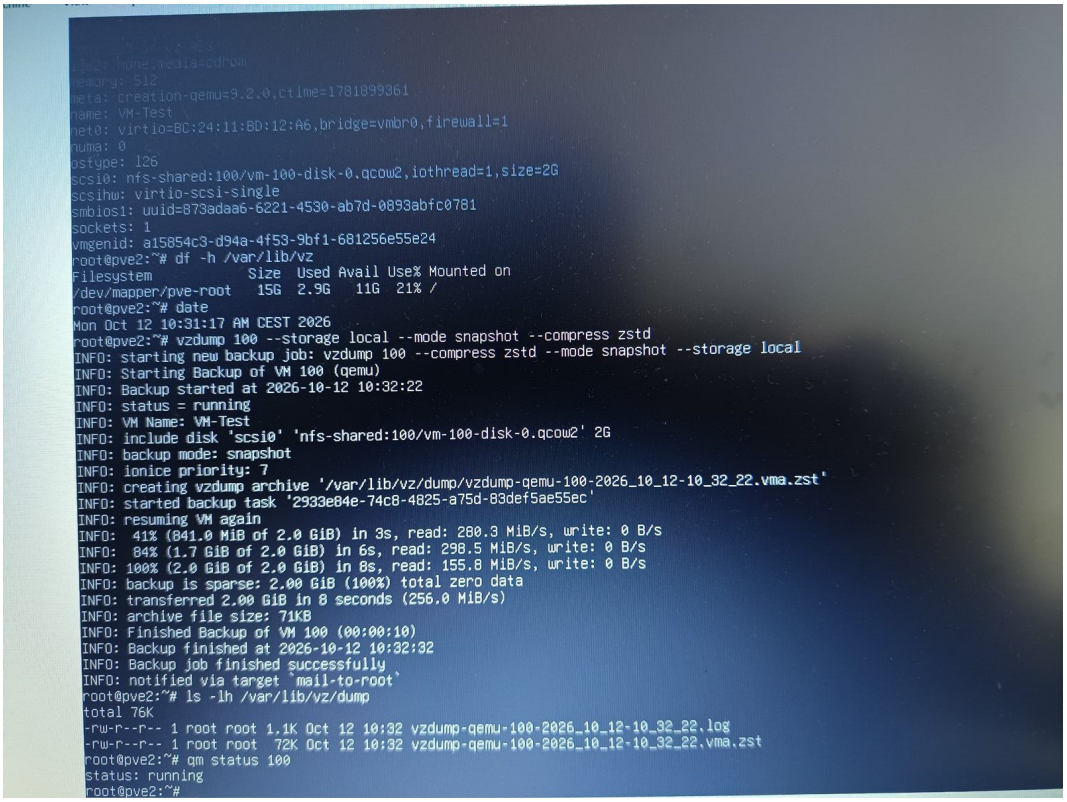


df -h /var/lib/vz on pve2: 11 GB available

1.2 — Manual backup

```
vzdump 100 --storage local --mode snapshot --compress zstd
```

Measurement	Value
Start / end	10:32:22 / 10:32:32
Duration	10 seconds
Archive size	72 KB (2 GB disk entirely empty: "sparse 100%")
Does the VM keep running?	Yes: snapshot mode, qm status 100 → running



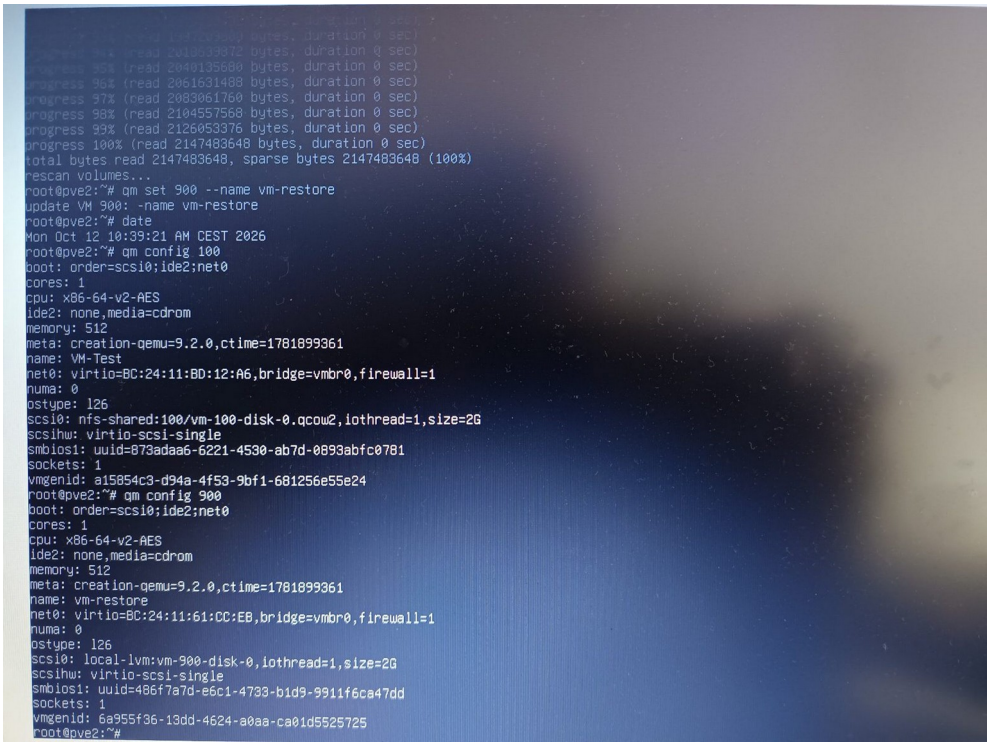
vzdump backup finished successfully, archive in /var/lib/vz/dump, VM still running

1.3 — Test restore as ID 900

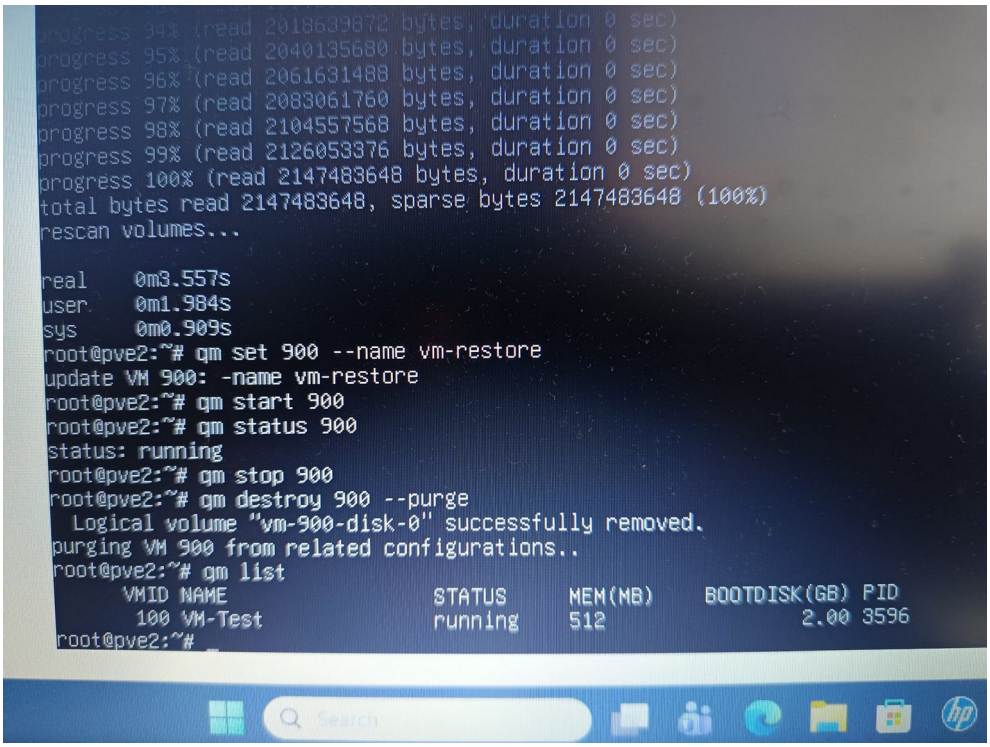
`qmrestore /var/lib/vz/dump/vzdump-qemu-100-2026_10_12-10_32_22.vma.zst 900 --storage local-lvm --unique 1`

The restore uses another ID (900), on local-lvm, with --unique (new MAC address), so as not to overwrite the running VM 100 or disturb HA. The first restore succeeded but its start time had scrolled off the screen; it was redone with the time command, which measures the exact duration.

	VM 100 (VM-Test)	**VM 900 (vm-restore)**
RAM / CPU	512 Mo / 1	identical
Disk	nfs-shared, 2 Go	local-lvm, 2 Go
MAC address	BC:24:11:BD:12:A6	BC:24:11:61:CC:EB (different)
Restore duration	—	3.6 s (real 0m3.557s)



Comparing qm config 100 / qm config 900



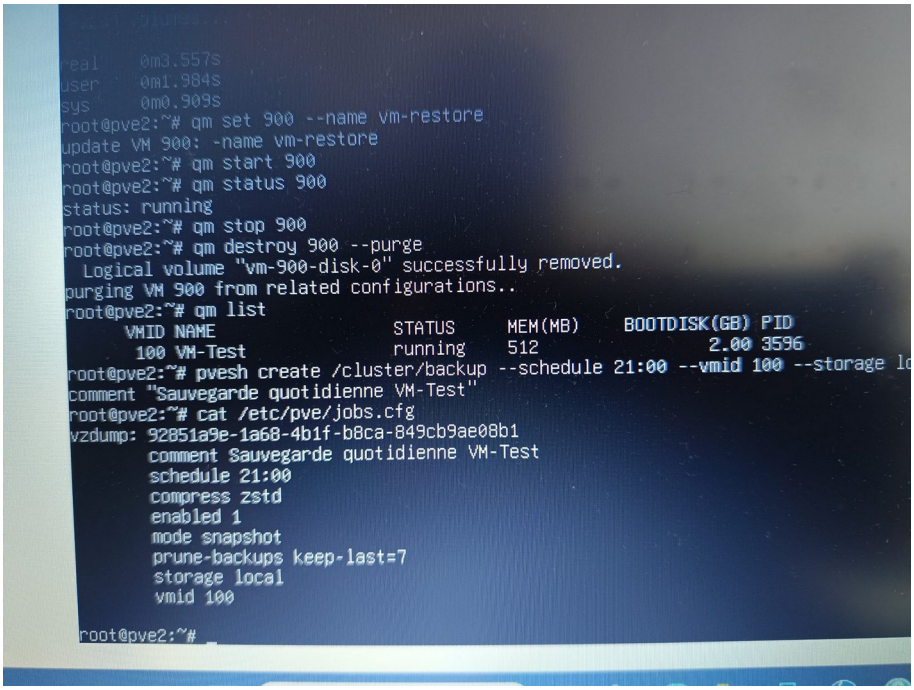
Timed restore (3.6 s), VM 900 started (running) then deleted; only VM 100 remains

VM-Test has no operating system: proof of start-up is the "running" state (the console would only show the BIOS). The 3.6 s duration is not representative of a real VM: a full 50 GB disk would take several minutes.

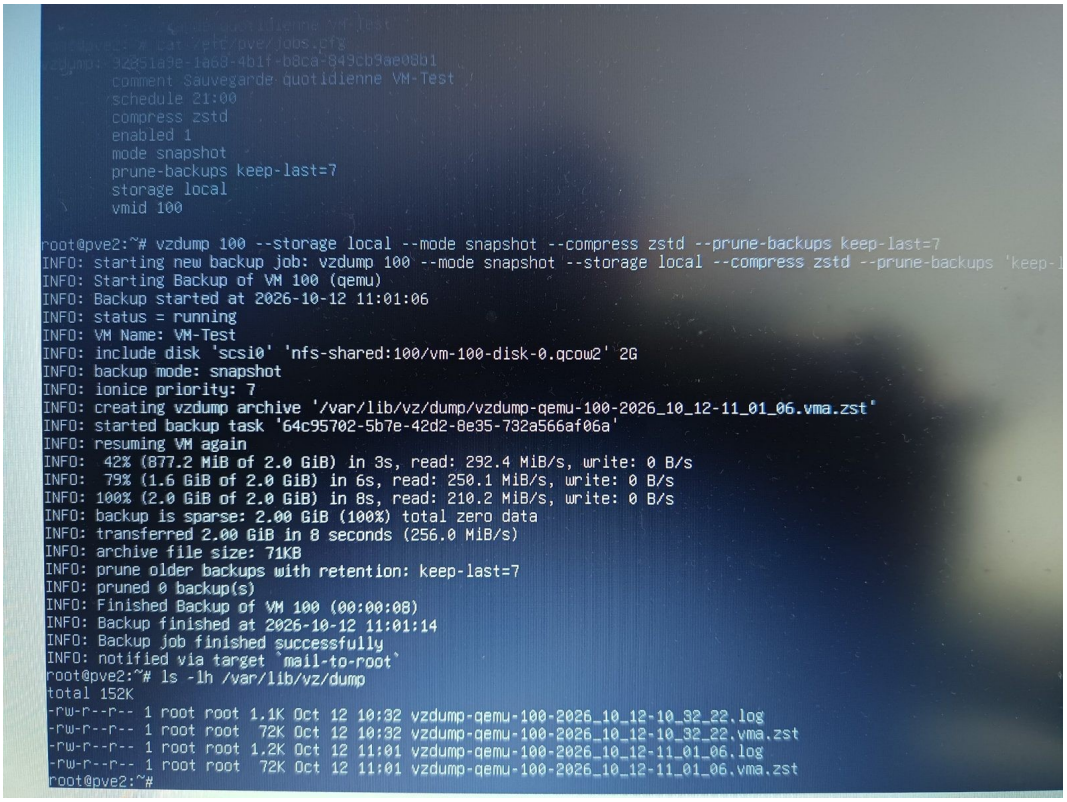
1.4 — Automatic backup

```
pvesh create /cluster/backup --schedule 21:00 --vmid 100 --storage local --mode snapshot --compress zstd --prune-backups keep-last=7 --enabled 1
```

Scheduled job every day at 21:00, snapshot mode, keeping 7 backups (older ones are deleted automatically). "Run now" was done by running vzdump with the same settings: backup in 8 s, "pruned 0 backup(s)" (nothing to delete, only 2 backups).



/etc/pve/jobs.cfg: schedule 21:00, prune-backups keep-last=7



"Run now": new backup at 11:01, 2 backups in local storage

1.5 — Does the lab follow the 3-2-1 rule?

Question	Answer
How many copies of the disk? Where?	3: the original on nfs-shared (pve's disk) and 2 backups in /var/lib/vz/dump on pve2
On how many physical media?	1 in reality: the 3 nodes are VMs on the same PC disk
An off-site copy?	No
If the PC burns, what is left?	Nothing
What would Santélia need?	A Proxmox Backup Server on separate hardware (2nd medium) and an off-site copy, ideally immutable (against ransomware), as planned in the requirements spec

Conclusion: 3 copies, but neither 2 media nor an off-site copy. The lab therefore does not follow the 3-2-1 rule.

PART 2 — RPO: how much data do we lose?

2.1 — RPO by scenario (backup every evening at 21:00)

Scenario	What saves you?	RPO (maximum loss)
The node running the VM fails	HA (disk on NFS shared storage)	≈ 0 (only data in memory not yet written)
VM deleted by mistake at 18:00	Last night's 21:00 backup (HA cannot help)	≈ 21 h
Ransomware encrypts the VM's data	The last clean backup, from before the infection	≥ 24 h, up to 7 days (retention)
pve1 (NFS server) destroyed	The backup stored on pve2 (HA cannot help: the shared disk is gone)	Up to 24 h

ZFS groups: replication every 15 min	ZFS replication + HA (not applicable: our group uses NFS)	Up to 15 min

HA protects against a node's hardware failure, not against deletion, ransomware or loss of the storage: in those cases only a backup lets you go back in time. Lab limit: the backups are on the same PC as the data.

2.2 — Practical measurement

Not carried out: VM-Test has no Linux system. The RPO was estimated (table 2.1), not measured, as the lab guide allows.

PART 3 — Santélia disaster recovery plan (DRP)

The document to open on the day of a disaster. It must let someone who did not build the infrastructure bring the service back.

1. Scope

- Services covered: online appointment booking for the 3 clinics (Évry, Corbeil-Essonnes, Ris-Orangis) and the patient records software, identified as critical in the requirements spec.
- Infrastructure: 3-node Proxmox cluster "labcluster" (pve, pve2, pve3), critical VM disks on nfs-shared shared storage, VMs protected by HA.
- Backups: every day at 21:00, 7 backups kept. In the lab, the appointment VM is represented by VM-Test (VM 100).
- Out of scope: workstations, telephony, medical equipment.

2. Disaster scenarios

Scenario	Solution	RTO	RPO	BCP or DRP?
Node failure	HA + NFS shared storage	2 min 17 s (measured)	≈ 0	BCP (PCA)
Storage server failure	Restore from backup on another node (local-lvm)	3.6 s measured (empty disk); estimated 15–60 min in production	≤ 24 h	DRP (PRA)
Accidental VM deletion	Restore from the latest backup	Same	≤ 24 h	DRP (PRA)
Ransomware	Isolate, then restore the last clean backup (ideally an immutable off-site copy)	Several hours (analysis + restore)	≥ 24 h, up to 7 days	DRP (PRA)
Fire in the Évry server room	Rebuild on another site from the off-site copy (to be set up)	Hours to days	≤ 24 h with a daily off-site copy	DRP (PRA)

3. Step-by-step restore procedure

1. Confirm the incident and alert the person who decides to trigger the DRP (section 4).
2. Log in as root on a working node and check the cluster: pvecm status → Quorate: Yes.
3. Find the backup: ls -lh /var/lib/vz/dump/ on the node that ran the VM (web interface: node > local > Backups). Pick the most recent vzdump-qemu-<id>-<date>.vma.zst from before the incident (ransomware: from before the infection).

4. Restore under a new ID, without overwriting the old VM: qmrestore <file> <new id> --storage local-lvm --unique 1 (web interface: Restore, Storage local-lvm, tick Unique).
5. Check the configuration: qm config <id> (RAM, CPU, disk).
6. Start it: qm start <id>, then qm status <id> → running. Open the console.
7. Check the service: open the appointment page, log in with a test account, create then cancel a test appointment.
8. Put the VM back under HA: ha-manager add vm:<id> --state started (and remove the old HA resource).
9. Announce that the service is back. Record start and end times (actual RTO) and the period of lost data (actual RPO) in the incident log.

4. Roles and responsibilities

Role	Who?	What they do
Decides to trigger the DRP	Santélia management (delegated to IT outside office hours)	Assesses the impact, approves the trigger and the accepted data loss
Carries out the restore	The IT team (2 people: a lead and a backup)	Follows the procedure, checks the service, records the times
Informs the clinics and patients	Communication / clinic reception lead, with management	Informs staff, organises phone booking as a fallback, contacts the patients affected

5. Targets compared with measurements

Indicator	Spec target	Measured in the lab	Compliant?
RTO (node failure)	Not set (to be confirmed)	2 min 17 s	To be confirmed
RTO (restore)	Not set	3.6 s (empty disk, not representative)	To be confirmed
RPO (node failure)	Not set	≈ 0	Yes
RPO (restore)	Not set	Up to 24 h	To be confirmed

The requirements spec leaves RTO and RPO "to be confirmed with the client". Proposal for management to approve: RTO ≤ 1 h for appointment booking; RPO ≤ 24 h for appointments and lower for patient records (more frequent backups or replication).

6. Testing and updating the plan

- Daily: automatic check of backup results, with an alert on failure (Zabbix or email).
- Monthly: test restore under another ID, as in Part 1.3, measuring the duration.
- Yearly, and before the HAS audit: full disaster exercise (simulated failure and restore).
- Updated by IT after every test, every real incident and every infrastructure change (new server, new VM, new provider), and at least once a year. Each version is dated and approved by management.

PART 4 — Cross-group presentation

	Group 1	Group 2	Group 3
Storage (NFS / ZFS)	To fill in	To fill in	NFS
Measured RTO			2 min 17 s (node failure)
Measured or estimated RPO			≈ 0 (HA); ≤ 24 h (backup)
Single point of failure			The NFS share hosted on pve (and the single PC)

Comprehension questions

1. HA is in place: why are backups still needed?

HA only protects against a node failure: it restarts the VM elsewhere, in the state it was in. It does not protect against a deleted VM, data encrypted by ransomware (the VM would restart with its encrypted data), corruption, loss of the NFS storage or a site disaster. Only a dated copy lets you go back in time.

2. Why restore the test backup under a different VM number?

So as not to overwrite VM 100, which is running and protected by HA, and to test the backup without interrupting the service. The Unique option gives a new MAC address to avoid a network conflict. If the backup is bad, the original stays intact.

3. Your backup is on the local storage of the node running the VM. What happens if that node is destroyed?

The backups disappear with the node. In our lab, if pve2 is destroyed, HA restarts VM-Test elsewhere (its disk is on NFS), but all the backups are lost: after a later ransomware attack or deletion, there would be nothing left to restore. A backup must live on separate storage (Proxmox Backup Server on another machine) with an off-site copy: the 3-2-1 rule.

4. For Santélia's appointment booking, is a low RPO or a low RTO more important? Explain.

A low RTO comes first: while the service is down, patients cannot book and the clinics' activity is blocked (the March outage lasted over 6 hours). Losing a few recent appointments can be recovered (calling patients back, confirmations sent by email or SMS), so an RPO of a few hours is acceptable. For patient records it is the opposite: medical data must not be lost, so a low RPO comes first.

Takeaways and lessons

- A backup is only worth something if its restore has been tested: here, a 10 s backup and a proven 3.6 s restore.
- HA (BCP) and backups (DRP) complement each other: one keeps the service running, the other lets you go back in time.
- The lab does not follow the 3-2-1 rule: everything is on one PC. In production, a separate backup server and an off-site copy are needed.
- Limited resources: close unneeded applications before starting the 3 nodes, and check ha-manager status after every start-up.